

Market Power and Contagion in Interbank Networks

An Agent-Based Model of Systemic Risk

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19 June 2026

Outline

- 1 Abstract
- 2 Risk and Connectivity
- 3 Model Framework
- 4 Interest Rate and Market Power
- 5 Equity and Bankruptcy Dynamics
- 6 Phase Transition and Results
- 7 Conclusions

Abstract: ABM, Interbank Market, and Contagion

- **Agent-Based Models** to study systemic risk in interbank credit networks.
- Relationship between **network connectivity**, **market power**, and **bankruptcy cascades**.
- **Non-monotonic trade-off** between risk sharing and systemic risk—connectivity diversifies risk in normal times but amplifies contagion during crises.
- The 2008 financial crisis illustrates how risk-sharing mechanisms can fail under high leverage and heterogeneous agents.

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The 2008 Crisis: Background

- Increased bank leverage and corporate debt-equity ratios made the economy fragile.
- Traditional theory does not fully account for complex agent interactions during crises.
- The interbank market became a **contagion vector**: failure of one bank triggers a domino effect through interlocking exposures.

The Risk-Sharing Trade-off

Two opposing effects of network connectivity:

- **Benefit:** Increasing connectivity allows **risk sharing**—banks diversify exposure across more counterparties.
- **Cost:** Excessive connectivity increases **systemic risk**—default propagates through the network.
- Evidence: a **non-monotonic relationship** exists—low connectivity stabilizes by diluting shocks, but high connectivity facilitates cascading failures.

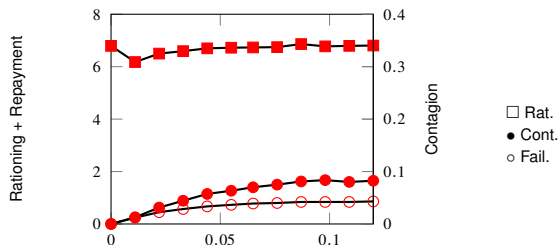
Channels of Contagion in Financial Crises

Three primary propagation mechanisms of systemic failure:

- 1 **Bank runs:** self-fulfilling panics where depositors withdraw mass funds simultaneously.
- 2 **Asset price contagion:** fire-sale spirals that depreciate assets across the system.
- 3 **Inter-locking exposures:** direct linkages through the interbank market—default of one bank triggers defaults in connected banks.

Model exposes that through **rationing** (borrower cannot secure sufficient funding), **contagion** (lender suffers losses from borrower default), and **repayment failure**.

Bankruptcies exposed by the model



p	Rat.	Cont.	Fail.
0.00001	6.79	0.00	0.00
0.011	6.18	0.01	0.24
0.022	6.50	0.03	0.45
0.033	6.59	0.04	0.57
0.044	6.70	0.06	0.67
0.055	6.72	0.06	0.73
0.065	6.73	0.07	0.78
0.076	6.75	0.07	0.80
0.087	6.86	0.08	0.84
0.098	6.77	0.08	0.84
0.109	6.79	0.08	0.84
0.120	6.81	0.08	0.86

	Assets	Liabilities
Initial conditions for each bank:	$C_0^i = 5 - 0.18 = 4.82$	$D_0^i = 9$
	$L_0^i = 5$	$E_0^i = 1$
	$R_0^i = 0.18$	

Monte Carlo simulations with $MC = 30$ runs, $N = 50$ banks, $T = 1000$ periods, and p varying from 0 to 0.12 in increments of 0.01.

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Model Overview

- ABM with N banks in a sequential economy $t = \{0, 1, \dots, T\}$.
- Banks connected through an **Erdős–Rényi** random graph $G(N, p)$: each edge exists with probability p .
- Three interacting sectors: goods market, credit market, interbank market.
- Goal: analyze how **connectivity** and **market power** affect system stability.

Balance Sheet

Accounting Identity

$$L_i^t + C_i^t + R_i^t = D_i^t + E_i^t$$

- **Assets:** Long-term loans (L), liquidity (C), reserves (R).
- **Liabilities:** Deposits (D), equity (E).
- Reserves set by regulation: $R_i^t = \hat{r}D_i^t$ with $\hat{r} = 0.02$.

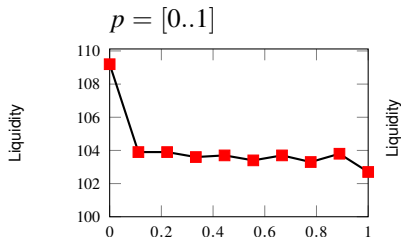
Liquidity Shocks

Deposits are exogenously shocked each period:

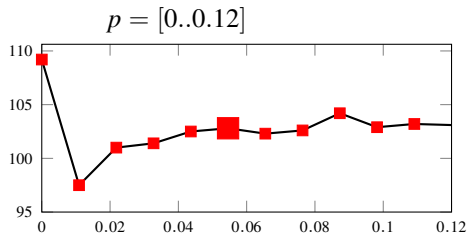
$$D_i^t = D_i^{t-1} [\mu + \omega \cdot U(0, 1)]$$

- μ : expected growth rate (0.7).
- ω : shock magnitude (0.35).
- **Potential lenders:** banks with positive net deposits (after reserves).
- **Potential borrowers:** banks with negative net deposits exceeding liquidity.

Liquidity in the model

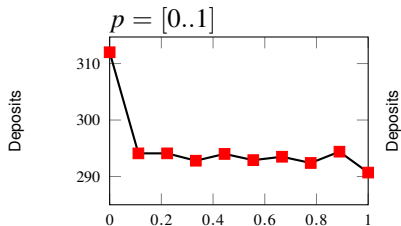


p	Val.
0.00001	109.2
0.111	103.9
0.222	103.9
0.333	103.6
0.444	103.7
0.556	103.4
0.667	103.7
0.778	103.3
0.889	103.8
1.000	102.7

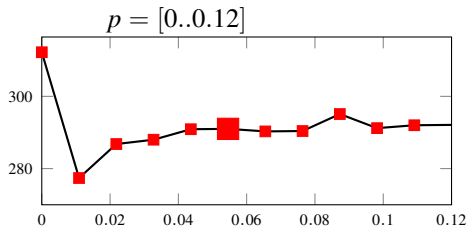


p	Val.
0.00001	109.2
0.011	97.5
0.022	101.0
0.033	101.4
0.044	102.5
0.055	102.8
0.065	102.3
0.076	102.6
0.087	104.2
0.098	102.9
0.109	103.2
0.120	103.1

Deposits in the model



p	Val.
0.00001	312.0
0.111	294.1
0.222	294.1
0.333	292.8
0.444	294.0
0.556	292.9
0.667	293.5
0.778	292.4
0.889	294.4
1.000	290.7

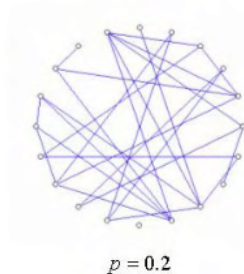
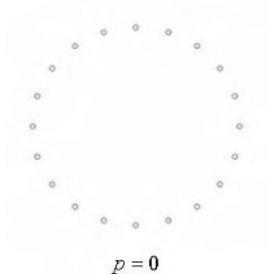


p	Val.
0.00001	312.2
0.011	277.4
0.022	286.8
0.033	288.0
0.044	290.9
0.055	291.0
0.065	290.3
0.076	290.4
0.087	295.1
0.098	291.2
0.109	292.0
0.120	292.1

Network Structure

- Bilateral credit lines formed at the start of each period.
- Connectivity controlled by parameter p in the Erdős–Rényi graph.
- Each borrower is matched to **exactly one** lender per period.
- The **Giant Component Size (GCS)** measures network integration.

Erdős–Rényi Random Graph



Each of the $\binom{N}{2}$ possible edges is included independently with probability p . For $N = 50$ banks:

- $p = 0$: isolated banks, no interbank market.
- $p = 1/N = 0.02$: **percolation threshold**—giant component emerges.
- $p > 0.05$: network is saturated ($> 92\%$ of banks connected).

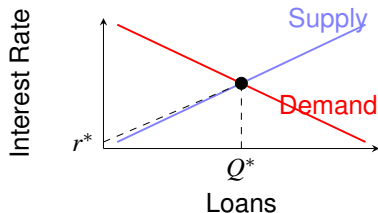
Network Structure

The Erdős–Rényi model does not attempt to replicate the entire interbank market; rather, it illustrates how connection density alters the probability and scale of contagion.

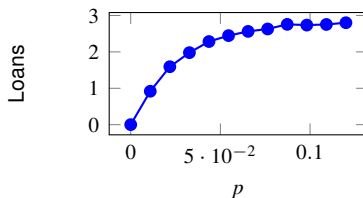
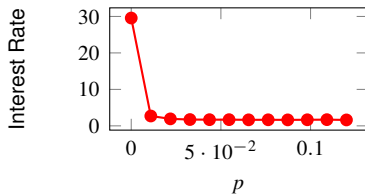
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Law of Supply and Demand



Model manages **market power** as a parameter ψ , which correlates with bank size and reflects the capacity to influence market terms, respecting the law of supply and demand.



Expected Profit

Banks are risk-neutral in perfect competition. Expected profit from lending to borrower j :

$$E[\Pi_{i,j}^t] = p_j^t(r_{i,j}^t - m)c_{i,j}^t - (1 - p_j^t)m \cdot c_{i,j}^t$$

- m : screening cost (0.015).
- p_j^t : probability of borrower survival.
- $r_{i,j}^t$: interest rate; $c_{i,j}^t$: lending capacity.

Interest Rate Derivation

Setting $E[\Pi] = 0$ (zero-profit condition):

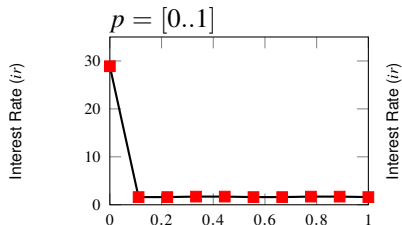
$$r_{i,j}^t = \frac{m}{p_j^t}$$

Introducing **market power** $\psi \in [0, 1)$:

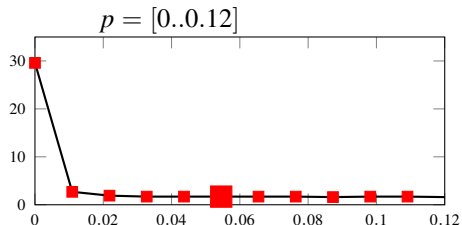
$$r_{i,j}^t = \frac{m}{p_j^t(1 - \psi)}$$

- $\psi = 0$: competitive (actuarially fair) rate.
- $\psi > 0$: monopolistic markup from information asymmetry.

Interest Rate in the model



p	Val.
0.00001	28.9
0.111	1.6
0.222	1.6
0.333	1.7
0.444	1.7
0.556	1.6
0.667	1.6
0.778	1.7
0.889	1.7
1.000	1.6



p	Val.
0.00001	29.6
0.011	2.7
0.022	1.9
0.033	1.7
0.044	1.7
0.055	1.7
0.065	1.7
0.076	1.7
0.087	1.6
0.098	1.7
0.109	1.7
0.120	1.6

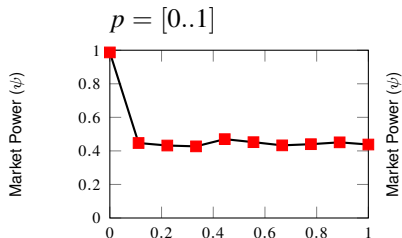
Endogenous Market Power

Market power evolves endogenously with bank size:

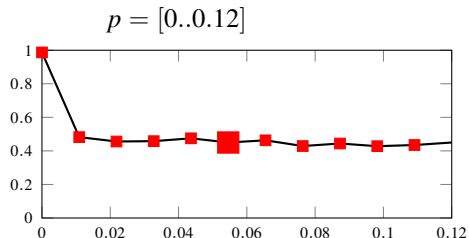
$$\psi_i^t = \frac{E_i^t}{E_i^{t,\max}}$$

- Larger banks (higher equity) exercise more market power.
- Borrower default probability: $p_{b,j}^t = E_j^t / E_j^{t,\max}$.
- Leverage: $\lambda_j^t = d_j^t / E_j^t$.
- The model evolves **only** from screening costs and shock parameters.

Market Power in the model



p	Val.
0.00001	0.987
0.111	0.447
0.222	0.432
0.333	0.427
0.444	0.470
0.556	0.452
0.667	0.433
0.778	0.440
0.889	0.451
1.000	0.438



p	Val.
0.00001	0.987
0.011	0.482
0.022	0.456
0.033	0.458
0.044	0.475
0.055	0.450
0.065	0.463
0.076	0.429
0.087	0.444
0.098	0.428
0.109	0.435
0.120	0.450

Outline

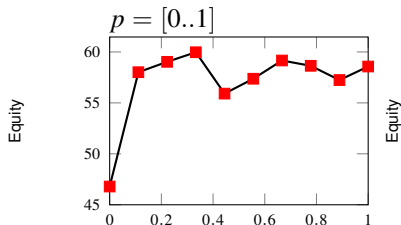
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Equity Evolution

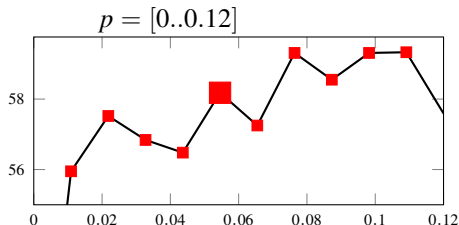
$$E_t^i = E_{t-1}^i + \sum_j l_t^{i,j} r_t^{i,j} - \sum_{j \subseteq \theta_t^i} [B_t^{i,j} - \tilde{L}_t^j]$$

- **Bankruptcy:** bank exits if $E_i^t < 0$.
- **Fire sales:** last-resort option (blocked in this model to amplify contagion effects).
- Failed banks replaced by new entrants with small initial capital.

Equity in the model



p	Val.
0.00001	46.79
0.111	58.01
0.222	59.03
0.333	59.97
0.444	55.91
0.556	57.37
0.667	59.16
0.778	58.64
0.889	57.24
1.000	58.57



p	Val.
0.00001	46.81
0.011	55.95
0.022	57.52
0.033	56.84
0.044	56.48
0.055	58.18
0.065	57.25
0.076	59.31
0.087	58.55
0.098	59.31
0.109	59.33
0.120	57.59

Failure Channels

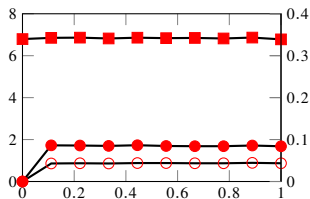
Three distinct causes of bank failure:

- 1 **Rationing:** borrower cannot secure sufficient funding.
- 2 **Contagion:** lender suffers losses from borrower default (bad debt).
- 3 **Repayment failure:** fully-funded borrower cannot repay loan or interest.

Key finding: **repayment failure dominates** at high connectivity, while rationing dominates at low connectivity. Contagion remains minimal ($< 5\%$).

Failures

$$p = [0..1]$$

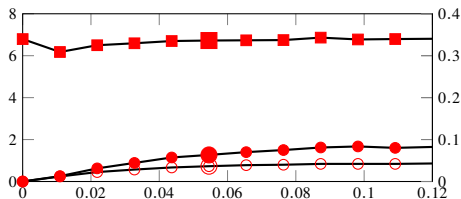


p Rat. Cont. Fail.

0.00001	6.79	0.00	0.00
0.111	6.85	0.09	0.86
0.222	6.86	0.09	0.87
0.333	6.82	0.08	0.86
0.444	6.86	0.09	0.88
0.556	6.84	0.08	0.88
0.667	6.85	0.08	0.87
0.778	6.82	0.08	0.87
0.889	6.86	0.09	0.89
1.000	6.78	0.08	0.87

■ Rationing
○ Failed
● Contagion

$$p = [0..0.12]$$



p Rat. Cont. Fail.

0.00001	6.79	0.00	0.00
0.011	6.18	0.01	0.24
0.022	6.50	0.03	0.45
0.033	6.59	0.04	0.57
0.044	6.70	0.06	0.67
0.055	6.72	0.06	0.73
0.065	6.73	0.07	0.78
0.076	6.75	0.07	0.80
0.087	6.86	0.08	0.84
0.098	6.77	0.08	0.84
0.109	6.79	0.08	0.84
0.120	6.81	0.08	0.86

■ Rationing
○ Failed
● Contagion

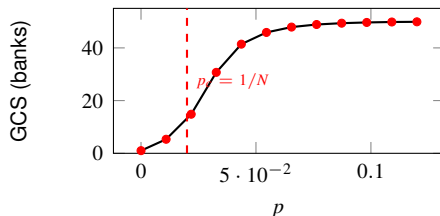
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Phase Transition

- **Percolation threshold:** GCS jumps from ~ 1 to ~ 46 banks (92%).
- **Low p :** rationing-dominated, high interest rates, few loans.
- **High p :** repayment-failure-dominated, low rates, saturated network.
- $p > 0.05$: all macroeconomic metrics stabilize.

Phase Transition



Percolation threshold for

Erdős–Rényi graph:

$$p_c = \frac{1}{N} = \frac{1}{50} = 0.02$$

p	GCS
0.00001	1
0.011	5
0.022	15
0.033	31
0.044	41
0.055	46
0.065	48
0.087	49
≥ 0.11	50

Connectivity and Systemic Risk

- As p increases, the interest rate drops from 30 (upper bound) to ~ 1.65 .
- Bankruptcies remain stable at ~ 7 – 7.5 (limited by model structure).
- Active loans saturate at ~ 13 (26% of banks).
- Equity jumps from 46.8 to ~ 55 and then plateaus.
- Bad debt grows rapidly before stabilizing at ~ 1.3 .

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Key Findings

As connectivity p increases:

- 1 The network becomes more connected.
- 2 Market power decreases.
- 3 The probability of bankruptcy rises.
- 4 The equilibrium interest rate falls.
- 5 Market power has a greater effect on the interest rate than the probability of bankruptcy does.
- 6 Competition mitigates risk more effectively than isolation.

Key Findings

- 1 **Phase transition at $p \approx 0.05$:** the network saturates and all metrics stabilize.
- 2 **Failure composition shifts:** rationing dominates at low p ; repayment failure dominates at high p .
- 3 **Contagion is minimal** ($< 5\%$ of failures)—the dominant risk channel is business failure, not network propagation.
- 4 **Market power** (ψ) is positively correlated with rates and decreases as connectivity increases.

Policy Implications

- **Heterogeneity** is a root cause of systemic instability; failure of large banks is catastrophic.
- Interbank lending should ideally be restricted among banks with **similar liquidity characteristics**.
- Reserve requirements dampen cascades but **limit growth**.
- Risk diversification across many agents has **ambiguous** effects on total systemic stability.